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B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Fourth Semester

21CSC206T - ARTIFICIAL INTELLIGENCE

(For the candidates admitted from the academic year 2021-2022 to 2023-2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

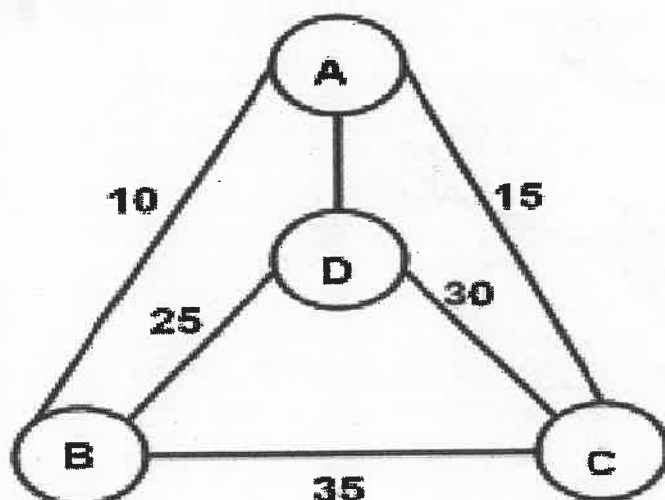
1. Which of the Following Sentence is TRUE 1 1 1 1
 - A) Problem solving is the process of generating solutions for a given situation
 - B) Problem will not be defined in a context manner.
 - C) Every problem wont have a well-defined objective.
 - D) Explore Information
2. What is the first step in data acquisition for AI models? 1 1 1 1
 - A) Data Annotation
 - B) Data Collection
 - C) Data Preprocessing
 - D) Data Augmentation
3. What is the objective of the Missionaries and Cannibals problem? 1 1 1 1
 - A) To ensure all missionaries cross before the cannibals
 - B) To get all missionaries and cannibals across the river without any being eaten
 - C) To prevent the cannibals from crossing the river
 - D) To make sure missionaries outnumber cannibals on both sides of the river
4. Which AI model is typically used for tasks like image classification? 1 1 1 1
 - A) Convolutional Neural Networks (CNNs)
 - B) K-nearest neighbors (KNN)
 - C) Support Vector Machines (SVM)
 - D) Recurrent Neural Networks (RNNs)
5. Unlike depth-first search and breadth-first search, Best First Search does not guarantee 1 1 2 1
 - A) Domain exploration
 - B) Agents
 - C) Completeness or optimality
 - D) Recurrent
6. Best First Search is an informed search algorithm that selects the node for expansion based on _____. 1 1 2 1
 - A) Heuristic function value
 - B) Hash value
 - C) algorithm
 - D) Process
7. _____ is the maximum number of edges in a binary tree with N nodes? 1 1 2 2
 - A) 5
 - B) 6
 - C) 7
 - D) n-1
8. The time complexity of Depth Limited Search is _____, similar to Depth First Search. 1 1 2 1
 - A) $O(b^d)$
 - B) C
 - C) D
 - D) $E*B$

9. Which of the following best defines a task environment in artificial intelligence (AI)? 1 1 3 2
 A) The hardware used to run AI algorithms B) The surrounding conditions in which an agent operates
 C) The goal set by the user D) The set of inputs used by the agent
10. Which of the following is a property of an intelligent agent? 1 2 3 2
 A) Adaptability B) Passivity
 C) Randomness D) Inactivity
11. Which of the following tasks is suitable for a multi-agent system? 1 2 3 1
 A) A single robot navigating a room B) Multiple autonomous drones coordinating to deliver packages
 C) A personal assistant answering queries D) A simple game of tic-tac-toe
12. What aspect of intelligent agents is most affected by flexibility? 1 2 3 1
 A) Stability B) Predictability
 C) Performance in dynamic environments D) Sensor accuracy
13. Which of the following statements is False about the Propositional Logic? 1 2 4 1
 A) Always involves quantifiers. B) Basis of symbolic reasoning systems
 C) Deals with truth values D) Uses logical connectives for statements
14. What type of reasoning is commonly used in knowledge-based agents to derive new information from existing knowledge? 1 1 4 2
 A) Deductive reasoning B) Inductive reasoning
 C) Abductive reasoning D) Heuristic reasoning
15. Which reasoning method is used to infer new facts from a set of rules and facts in the presence of uncertainty? 1 2 4 2
 A) Deductive reasoning B) Inductive reasoning
 C) Abductive reasoning D) Non-monotonic reasoning
16. What is the primary purpose of the unification algorithm in AI? 1 2 4 2
 A) To find the most efficient algorithm B) To match different logical expressions
 C) To optimize computational resources D) To enhance data storage
17. In Means-Ends Analysis, how is the current state defined? 1 2 5 2
 A) The desired final outcome of the problem B) The initial situation or configuration before any actions are taken
 C) Any possible state that can be reached D) A temporary state achieved through random actions
18. In which scenario is Means-Ends Analysis particularly useful? 1 2 5 2
 A) When the solution space is large and complex B) When a clear set of rules is available
 C) When there is only one solution D) When data is perfectly organized
19. Which of the following is a common approach to solve the Blocks World Problem? 1 2 5 3
 A) Random search B) Depth-first search
 C) Genetic algorithms D) Dynamic programming
20. What does forward chaining use to infer new facts? 1 1 5 2
 A) Rules and goals B) Known facts and rules
 C) Only goals D) Process

PART - B (5 x 8 = 40 Marks)
Answer ALL Questions

Marks BL CO PO

- 21 a. Analyse knowledge-based complexity for the AI application for travelling salesman problem 8 1 1 1



(OR)

- b. Describe about Problem Solving Process.

8 1 1 1

- 22 a. Explain the concept of a queue in data structures. Describe its main operations and give an example of a real-world application.

8 1 2 1

(OR)

- b. Explain Briefly about Depth First Search with suitable example

8 1 2 1

- 23 a. Explain the concept of performance measures in the context of intelligent agents. Discuss how performance measures relate to the rationality of an agent

8 2 3 1

(OR)

- b. Define "Task Environment" in the context of intelligent agents. Describe the properties that characterize a task environment.

8 1 3 1

- 24 a. Define inference in the context of artificial intelligence. Discuss the different types of inference methods used in AI.

8 1 4 2

(OR)

- b. Explain the Unification Algorithm in the context of logic programming. Discuss its importance, working mechanism, and provide examples to illustrate its application.

8 1 4 2

- 25 a. Explain Means-Ends Analysis in Artificial Intelligence. Discuss its role in problem-solving and learning, and provide examples of how it can be applied in real-world scenarios.

8 1 5 3

(OR)

- b. Explain the fundamental concepts of machine learning, and describe its key methods and models.

8 1 5 3

PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Question

Marks BL CO PO

26. Explain the concepts of Trees and Graphs. Discuss their types and key properties in real-world applications. Provide examples to illustrate their usage in solving complex problems.

15 1 2 2

27. Given the graph below, demonstrate how the A* algorithm searches for the shortest path from node A to node J. In your explanation, highlight how A* combines the advantages of Breadth First Search (BFS) and Best First Search (BFS) through its use of heuristics. Also, discuss the importance of the chosen heuristic being both admissible and consistent.

